

Dedicated Hardware Design for Chroma-from-Luma Intra Prediction in AV1 Decoders

Abstract—Chroma From Luma (CfL) is an intra prediction tool for chroma samples introduced in the AV1 video codec. Chroma samples are derived from collocated reconstructed luma samples through a linear model. Due to the significantly higher encoding time compared to AVC, HEVC, and VP9, hardware-based approaches have been explored to accelerate specific coding tools. This paper presents hardware architectures for the AV1 CfL tool, supporting all defined block sizes and two subsampling modes (4:2:0 and 4:2:2). We propose two architectures, one for single-sample input and the other with multi-sample inputs. The proposed design for the multi-sample input version achieves a clock frequency of up to 480 MHz, with a power consumption of 68.98 mW. Furthermore, it is capable of processing Full HD video for both subsampling modes.

Index Terms—AV1, Chroma From Luma, Hardware, Accelerator, VLSI.

I. INTRODUCTION

The increasing demand for high-resolution and high-quality video services, such as Ultra-High Definition (UHD) and High Dynamic Range (HDR), as well as large-scale internet streaming, has intensified the need for more efficient video compression standards. In response to these challenges, the Alliance for Open Media introduced the AOMedia Video 1 (AV1) codec, [1] as a next-generation, royalty-free solution designed to improve compression efficiency while supporting a wide range of modern video applications. AV1 achieves significant bitrate reductions compared to previous standards. When compared to VP9, AV1 offers more than 30% bitrate reduction [2]. AV1 obtains better Peak Signal-to-Noise Ratio (PSNR) results than Advanced Video Coding (H.264/AVC), High Efficiency Video Coding (H.265/HEVC), and VP9 for the same video bitrates and default parameters [3].

The AV1 encoder incorporates numerous advanced tools to achieve high compression efficiency, while also building upon methods inherited from earlier codecs (VP9, VP10, Daala, and Thor) [4]. Among these new tools, AV1 presents a more flexible block partitioning structure, a richer set of intra and inter prediction modes, and advanced in-loop filters. However, besides this performance increase, when compared to AVC, HEVC, and VP9, the encoding time is significantly higher [3]. In order to solve this problem, software-based complexity reduction solutions and hardware accelerators are used.

One of these new AV1 features is the Intra Prediction Chroma From Luma (CfL) mode [5]. CfL is an intra prediction mode used exclusively for chroma coding blocks. In this approach, chroma prediction samples are generated from the corresponding reconstructed luma samples using a linear model. AV1 was the first codec to present a method to predict

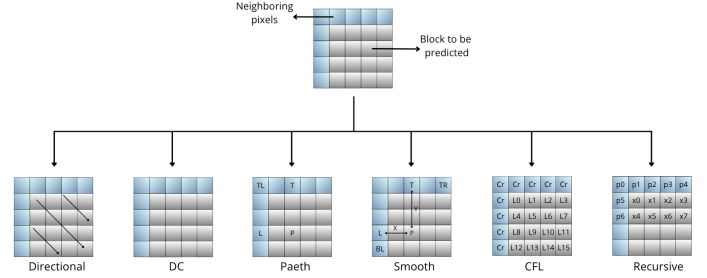


Fig. 1. AV1 Main Intra Prediction modes.

chroma pixels from luma information. According to [5], CfL is able to reduce the BD-rate by 4.87% for still images and 2.41% for video sequences, when measured with CIEDE2000.

This paper proposes a dedicated hardware design for the AV1 Decoder CfL mode. The proposed design supports all the block sizes available in AV1, two subsampling modes (4:2:0 and 4:2:2), and a channel depth of 10 bits. The design explores parallel architecture with the aim of supporting high resolutions for real-time processing. To the best of the author's knowledge, only one work in the literature proposes a hardware design for CfL mode. In [6], an architecture that supports only the 4:2:0 subsampling mode is proposed. In this work, the proposed architecture supports two subsampling modes (4:2:0 and 4:2:2) and aims to achieve higher clock frequencies and parallelism.

II. BACKGROUND

A. Intra Prediction in AV1

Intra prediction is a spatial prediction technique used in video codecs to exploit redundancy within a single frame. Instead of relying on information from previously and/or subsequently coded frames (Inter Prediction), intra prediction estimates the current block's samples using already reconstructed neighboring samples from the same frame, typically from the top and left boundaries. By modeling local spatial correlations such as edges, textures, and smooth regions, the encoder generates a predicted block and transmits only the residual (the difference between the original and predicted samples), along with the selected prediction mode. This process significantly reduces spatial redundancy and improves compression efficiency while maintaining visual quality.

In AV1, intra prediction comprises a rich set of directional and non-directional modes designed to efficiently exploit spatial redundancies within a frame, as shown in Fig. 1. The

codec supports 56 directional predictors, corresponding to angles ranging from 36 to 212 degrees, enabling accurate modeling of textures and edges with different orientations. In addition to directional modes, AV1 provides several non-directional predictors. These include: DC prediction, which estimates the block using the mean of neighboring samples; Paeth prediction, which selects the most suitable reference sample based on a simple linear model; Smooth prediction that perform linear interpolation between top and left reference boundaries; and Chroma-from-Luma (CfL), which is the focus of this paper. Additionally, it includes a recursive mode that models spatial correlation as a 2D first-order Markov process, and Palette mode, which represents a block using a small set of colors and an index map for screen content videos.

B. Chroma From Luma

Fig. 2 shows the CfL algorithm in the AV1 decoder. The CfL algorithm starts with the subsampling process of the reconstructed luma samples. Since chroma samples are subsampled according to the 4:2:0 or 4:2:2 formats, the luma samples must be resized to match their dimensions. In the 4:2:0 format, both the block width and height are reduced to half of their original dimensions. In the 4:2:2 format, only the block width is halved, while the block height remains unchanged. Subsequently, the DC component is removed to obtain the AC contribution. This operation is performed by subtracting the average value from the sample value. To approximate the chroma AC component from the reconstructed luma AC contribution, AV1-CfL encoder determines the scaling parameter using the original chroma samples and explicitly signals it in the bitstream, rather than requiring the decoder to infer it as in some prior approaches. This strategy reduces decoder complexity while enabling more accurate predictions. Finally, the DC chroma prediction is added to the result. The DC chroma prediction is calculated by the average operation of the left and/or above samples of the chroma block. Although AV1 supports 19 block-size combinations for intra ranging from 4×4 to 64×64 , the CfL design accepts only blocks with sizes smaller than or equal to 32×32 [7].

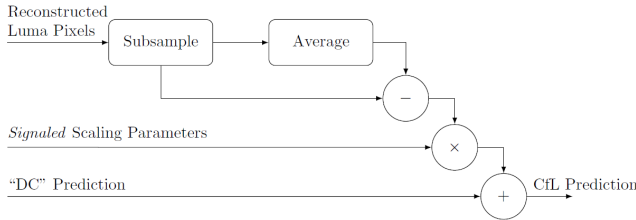


Fig. 2. CfL Algorithm block diagram in AV1 Decoder [5].

III. PROPOSED ARCHITECTURE

Fig. 3 presents our proposed hardware architecture. The architecture is able to process all the block sizes supported by AV1 for CfL. It also supports two subsampling modes: 4:2:0 and 4:2:2. The architecture is composed by three main blocks:

Chroma DC Prediction Processor, CfL Prediction Processor and Control Unit. Each processor is represented by a top-level block that contains its corresponding functional sub-blocks, depicted in blue. The Chroma DC Prediction Processor performs the calculation of the Chroma DC prediction by loading the chroma samples, adding of all them, performing a round operation and then the division. The result is sent to CfL Prediction Processor to be added to the CfL prediction. In the CfL Prediction Processor, the reconstructed luma samples are loaded and subsampled. Subsequently, the average of the subsampled samples are calculated and then, the AC calculation is performed by the subtraction of the average from the subsampled samples. In sequence, the scaling parameter is applied and finally, the result is added to the Chroma DC prediction value. Both processors perform its operations concurrently. The Control Unit controls the data path and it is connected to both processors.

A. DC Processor

The DC processor performs the DC prediction using chroma samples. This chroma prediction is required, as it is later combined with the scaled luma AC contribution in the CfL prediction algorithm. The proposed design for the DC processor is shown in Fig. 3b. Initially, the neighboring left and top chroma samples of the block in analysis are serially loaded into the Sum Left and Sum Top blocks, respectively. The Sum Left and Sum Top blocks are implemented as four-stage registered adder trees. The first-stage register stores 16 samples of up to 10 bits, with one sample loaded per clock cycle. In each stage, the samples are added pairwise, progressively reducing the number of operands until the final sum of the 16 samples is generated. These blocks are able to support all AV1 numbers of left and top samples: 4, 8, 16, 32 and 64. If the number of samples is less than 16, fewer stages are required to perform the summation. However, if the number of samples exceeds 16, partial sums are accumulated and the process restarts until the final sum is completed. Once the sums of the left and top samples are computed, the results are added together. Additionally, a value derived from the number of samples is also added for rounding purposes. Afterward, the result is divided by all the possible combinations of top and left sample numbers, using bit-shift operations. Finally, the DC prediction value is selected through a multiplexer according to the number of samples and the top/left selection signals. Although Fig. 3b illustrates the DC prediction process when both top and left neighboring samples are available, the proposed design also supports scenarios in which only left or only top samples are present.

B. CfL Processor

The CfL processor performs the CfL prediction using the reconstructed luma samples and the chroma DC prediction. The proposed architecture of the CfL processor is illustrated in Fig. 3c. Initially, the reconstructed luma samples of the block under analysis are serially loaded into the Luma Samples Memory, with one 10-bit sample being stored per clock cycle.

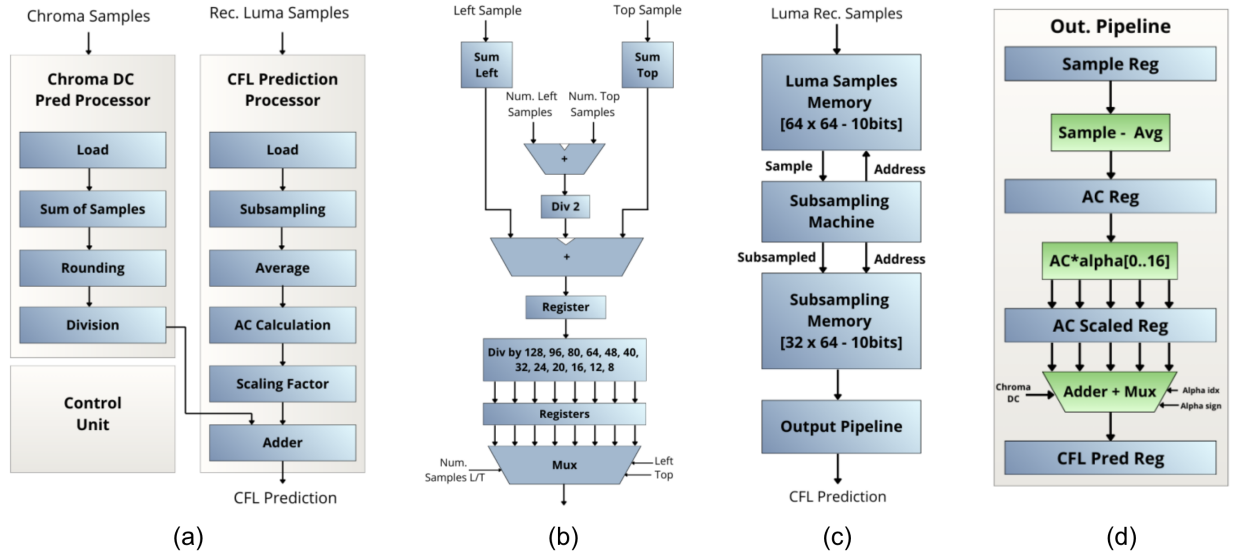


Fig. 3. (a) Top level architecture of the CfL unit, (b) DC processor architecture, (c) Proposed CFL processor architecture, and (d) CFL pipeline.

The Luma Samples Memory is implemented as a register file capable of storing up to 64×64 addresses of 10-bit samples. Once the memory is loaded according to the number of samples of the block under analysis, it signals the Subsampling Machine. This unit performs the subsampling operation and supports two subsampling modes: 4:2:2 and 4:2:0. To accelerate the process, four samples are processed simultaneously. The unit is implemented as a seven-state Finite State Machine (FSM), responsible for generating the Luma Memory addresses, loading the samples, computing the averages, generating the Subsampling Memory addresses, and writing the results to the Subsampling Memory. The Subsampling Memory is implemented as a register file capable of storing up to 32×64 addresses of 10-bit samples. Once the subsampling process is completed and the Subsampling Memory is loaded, the output pipeline phase begins. The Output Pipeline architecture is illustrated in Fig. 3d and is implemented as a three-stage pipeline. The first stage performs the AC component calculation, while the second stage computes the multiplication by the scaling parameter. The architecture supports 17 possible alpha values, each represented by an index. The values range from 0 to 2 with increments of 0.125 between consecutive indices. Additionally, for each alpha value, both positive and negative signs are supported. The AC value is multiplied by all possible alpha and sign combinations. In the final stage, the scaled AC value is selected according to the signaled alpha index and sign, and then added to the chroma DC value, generating the final CfL prediction. The output of the architecture is the CfL prediction value for the block analyzed.

C. Control Unit

The Control Unit is responsible for controlling the sequence of operations of the processors. It establishes synchronization between the DC and CfL processors and manages the execution order of specific operations, such as determining when

the subsampling process begins and when the output pipeline operation starts.

D. Multi-Sample Input Version

The previously described design was developed with a reduced number of I/O pins. For instance, the input luma samples were limited to a throughput of one 10-bit sample per clock cycle. Similarly, the chroma samples (top and left) were also restricted to 10-bit inputs. While this approach reduces the number of required pins, it introduces a significant latency during the sample loading process. To address this issue, a new version was implemented with increased loading throughput. The luma input bus was expanded to support 16 samples of 10 bits per clock cycle, enabling the loading of a 4×4 block per cycle. In the previous version, loading a 64×64 luma block required 4096 clock cycles, whereas the new design reduces this to 256 cycles. For the chroma input bus, both top and left channels were expanded to support four 10-bit samples per clock cycle. As a result, loading 32 top and 32 left chroma samples, which previously required 32 clock cycles, now takes only eight cycles.

IV. RESULTS

The proposed architecture was developed using Hardware Description Languages (HDL) such as Verilog and SystemVerilog. It was synthesized using Generic PDK for 45nm (GPDK045), by Cadence Design Systems. Table I presents the results of logic synthesis for both versions.

Although the multi-sample design increases the number of input pins, it reduces the overall area and power consumption due to a more parallel architecture, which minimizes control logic. Table II shows the clock cycle number required to process a single block for all blocks considering 4:2:0 subsampling. It can be seen that the number of required clock cycles

TABLE I
COMPARISON BETWEEN SINGLE-SAMPLE AND MULTI-SAMPLE VERSIONS

	Single-Sample Version	Multi-Sample Version
Clock Period	2.1 ns	2.08 ns
Num. Pins	75	285
Area	795471.548 μm^2	716040.065 μm^2
Cell Count	318066	278645
Power	76.25 mW	68.98 mW

dropped around 40.7% for 4:2:0 subsampling when using the Multi-Sample version.

In order to verify the performance of the design, the minimum frequency required to drive a video with 1920x1080 (1080p) resolution using the Multi-Sample Input version was calculated. Table III shows the results considering 4:2:0 and 4:2:2 subsampling modes. The calculation was performed considering a video frame divided into 8x8, 16x16, and 32x32 blocks. These block sizes were considered since they are the block sizes that occur the most among the non-directional intra modes for the luminance blocks [8]. The equation 1 was used to calculate the frequency values.

$$f = \frac{WH}{wh} \cdot \text{fps} \cdot N_{\text{cycles}} \quad (1)$$

Considering Table III, the design can handle Full High Definition (FHD) resolution at 40fps for 4:2:0 subsampling and 60fps for 4:2:0 subsampling. The current architecture is not capable of handling 4K resolution under the same conditions. At the synthesized frequency, approximately 1950 clock cycles per block would be required to sustain 4K video at 30fps using 4:2:0 subsampling mode and block size of 32x32. This is a cycle reduction of 63.8%. The main bottleneck is the subsampling process. For example, for the 4:2:0 subsampling and 32x32 block size, the subsampling process takes around 76% of the total number of cycles to process the block. Also, the output data bus is limited to one 10-bit sample, lowering the output throughput.

In comparison with [6], one key distinction is that the proposed design supports both 4:2:2 and 4:2:0 subsampling modes, whereas the design presented in [6] supports only the 4:2:0 mode. Another notable difference lies in the operating

TABLE II
CLOCK CYCLES COMPARISON BETWEEN SINGLE-SAMPLE AND MULTI-SAMPLE VERSIONS IN 4:2:0 SUBSAMPLING

DC Block Size	CFL Block Size	Clock Cycles (Single)	Clock Cycles (Multi)	Cycle Reduction (%)
4x4	8x8	156	97	37.82
8x8	16x16	588	349	40.65
16x16	32x32	2316	1357	41.41
32x32	64x64	9228	5389	41.60
4x8	8x16	299	180	39.80
8x4	16x8	299	180	39.80
8x16	16x32	1163	684	41.19
16x8	32x16	1163	684	41.19
16x32	32x64	4619	2700	41.55
32x16	64x32	4619	2700	41.55
4x16	8x32	587	348	40.72
16x4	32x8	587	348	40.72
8x32	16x64	2315	1356	41.43
32x8	64x16	2315	1356	41.43

TABLE III
MINIMUM FREQUENCY TO PROCESS FHD FOR 4:2:0 AND 4:2:2 SUBSAMPLING

Block Size	4:2:2 subsampling			4:2:0 subsampling		
	Res.@FPS	Cycles	Freq. (MHz)	Res.@FPS	Cycles	Freq. (MHz)
8x8	1080p@60	245	476.28	1080p@40	349	452.3
16x16	1080p@60	941	457.32	1080p@40	1357	439.6
32x32	1080p@60	3725	452.58	1080p@40	5389	436.5

frequency: the proposed architecture achieves up to 480 MHz, while the design in [6] operates at 132.1 MHz. On the other hand, despite its lower clock frequency, the [6] design is capable of supporting 4K resolution, whereas the proposed design is limited to FHD. This limitation arises from the bottleneck in the subsampling process required to support both 4:2:2 and 4:2:0 modes, which constitutes the main target for future improvements.

V. CONCLUSION

In this paper, a hardware architecture for the AV1 CfL decoder is presented. The proposed design supports 10-bit input samples, all block sizes defined for AV1 CfL, and both 4:2:0 and 4:2:2 subsampling modes. The design was synthesized using the Generic 45 nm PDK (GPD045) from Cadence Design Systems, achieving a clock frequency of 480 MHz. The power consumption is 68.98 mW for the Multi-Sample Input version and 76.25 mW for the Single-Sample Input version. The design is capable of achieving FHD resolution for both subsampling modes. It was also observed that the subsampling stage represents the main bottleneck of the system. Future work will focus on optimizing the Subsampling Machine block to reduce the number of clock cycles required for block processing. Additionally, the authors intend to expand the output bus. Both improvements aim to enable support for 4K resolution. Once the optimization is completed, the authors intend to perform physical synthesis.

REFERENCES

- [1] AOM, "AV1 video codec," <https://aomedia.org/specifications/av1/>, 2018.
- [2] J. Han, B. Li, D. Mukherjee, C.-H. Chiang, A. Grange, C. Chen, H. Su, S. Parker, S. Deng, U. Joshi, Y. Chen, Y. Wang, P. Wilkins, Y. Xu, and J. Bankoski, "A Technical Overview of AV1," *Proceedings of the IEEE*, vol. 109, no. 9, pp. 1435–1462, 2021.
- [3] M. A. Layek, N. Q. Thai, M. A. Hossain, N. T. Thu, L. P. Tuyen, A. Talukder, T. Chung, and E.-N. Huh, "Performance analysis of H.264, H.265, VP9 and AV1 video encoders," in *2017 19th Asia-Pacific Network Operations and Management Symposium (APNOMS)*, 2017, pp. 322–325.
- [4] Siqueira, G. Corrêa, and M. Grellert, "Complexity and coding efficiency assessment of aomedia video 1," in *2023 IEEE 14th Latin America Symposium on Circuits and Systems (LASCAS)*, 2023, pp. 1–4.
- [5] L. Trudeau et al., "Predicting chroma from luma in AV1," in *Data Compression Conference (DCC)*, Mar. 2018.
- [6] J. Goebel, B. Zatt, L. Agostini, and M. Porto, "Hardware design of DC/CFL intra-prediction decoder for the av1 codec," in *2019 32nd Symposium on Integrated Circuits and Systems Design (SBCCI)*, 2019, pp. 1–6.
- [7] Alliance for Open Media, "Aomedia av1 codec library (libaom) reference implementation," <https://aomedia.googleusercontent.com/aom/>, 2026, accessed: 2026-03-10.
- [8] M. M. Corrêa, B. H. Waskow, J. W. Goebel, D. M. Palomino, G. R. Corrêa, and L. V. Agostini, "A high-throughput hardware architecture for av1 non-directional intra modes," *IEEE Transactions on Circuits and Systems I: Regular Papers*, vol. 67, no. 5, pp. 1481–1494, 2020.